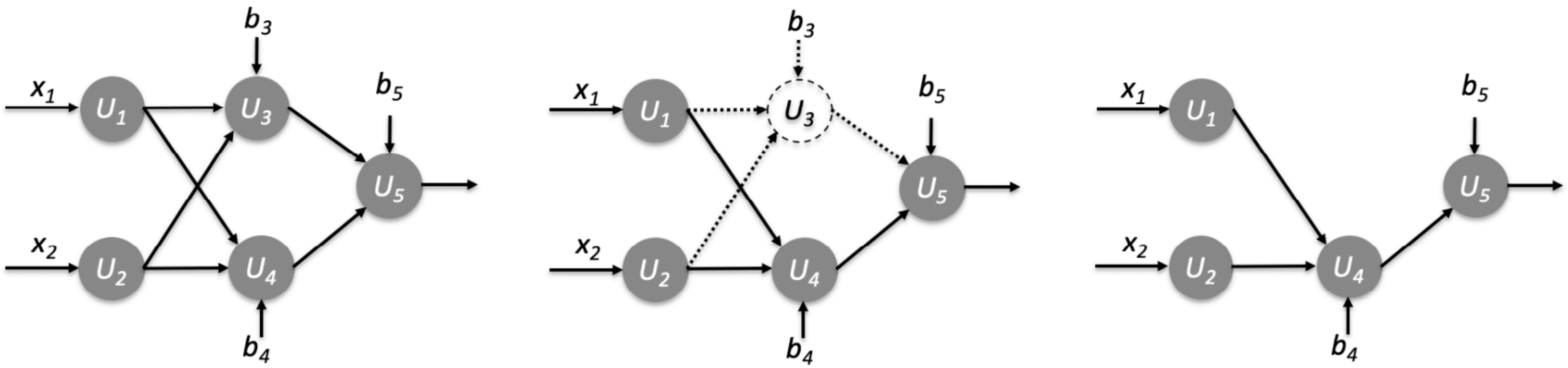
Dropout

- **The Purpose of Dropout: to Prevent Overfitting**
- **How does it works?**
  - At each epoch (ρ dropout rate, e.g., $\rho = 0.5$)
    - randomly dropout some non-output units
    - Perform B.P. on the dropout network
  - Scale the final model parameters $\boldsymbol{\theta}^* \leftarrow \rho \cdot \boldsymbol{\theta}^*$



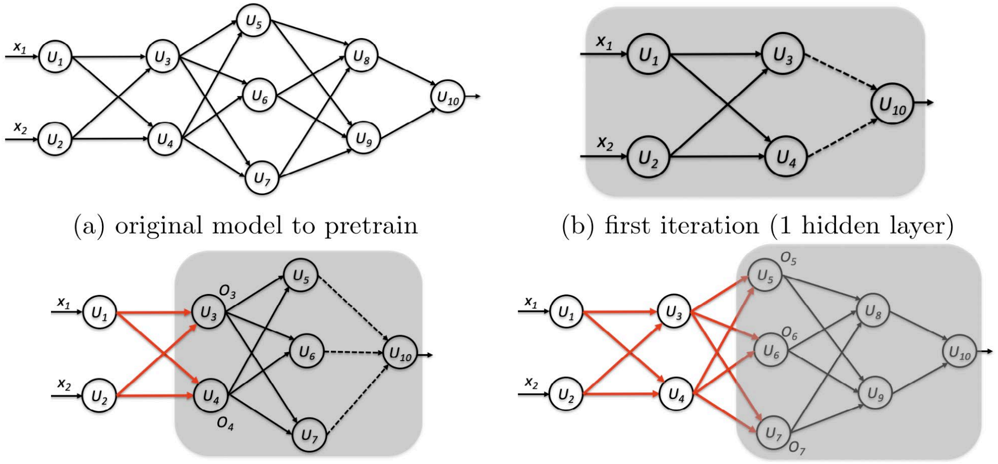
(a) Original network (b) Dropout U_3 (c) Dropout network

Why does Dropout Work?

- Dropout can be viewed as a regularization technique
  - Force the model to be more robust to noise, and to learn more generalizable features
- Dropout vs. Bagging
  - Bagging: Each base model is trained *independently* on a bootstrap sample
  - Dropout: the model parameters of the current dropout network are updated based on that of the previous dropout network(s)

Pretraining

- **Pretraining:** the process of initializing the model in a suitable region
- **Greedy supervised pretraining**
  - pre-set the model parameters layer-by-layer in a greedy way
  - Start with simple model, add one additional layer at a time,
  - pretrain the parameters of the newly added layer, while fixing for other layers
  - Each time, equivalent to training a two-layered network
  - Followed up a fine-tuning process
- **Other Pretraining Strategies**
  - Unsupervised pretraining: based on auto-encoder
  - Hybrid strategy
- **Pretraining for transfer learning**

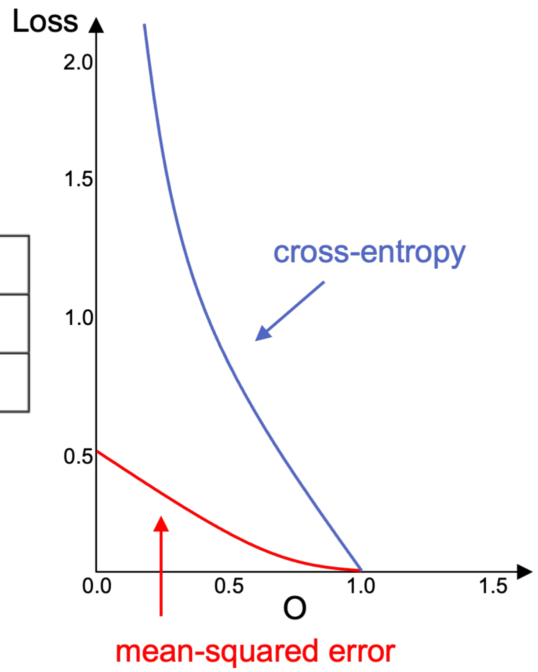


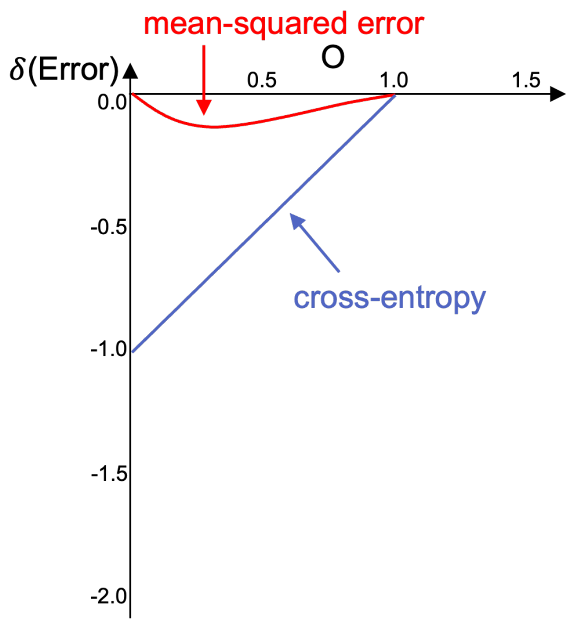
(a) original model to pretrain
(b) first iteration (1 hidden layer)
(c) second iteration (2 hidden layers)
(d) third iteration (3 hidden layers)

Cross Entropy

- **Measure the disagreement between the actual (T) and predicted (O) target values**
  - For regression: mean-squared error
  - For (binary) classification: cross-entropy
- **Mean-squared error vs. Cross-entropy**

	Mean-squared error	Cross entropy
Loss	$\frac{1}{2}(T-O)^2$	$-T\log O-(1-T)\log(1-O)$
Error δ	$O(1-O)(O-T)$	$O-T$





Assume positive example (*T=1*)

- **Cross-entropy for Multiclass Problem**
  - Actual target $T=(T_1,T_2,\ldots,T_C)$
  - Predicted output $O=(O_1,O_2,\ldots,O_C)$

$$\text{Loss}(T,O)=-\sum_{j=1}^{C}T_j\log O_j$$